IS SPIKE-DRIVEN SELF-ATTENTION NECESSARY? THE INEFFICIENCY OF JACCARD ATTENTION IN SPIKING SENTENCE EMBEDDINGS

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ABSTRACT

Spiking Neural Networks (SNNs) provide an energy-efficient alternative to conventional Deep Neural Networks by utilizing discrete, event-driven temporal spikes. However, adopting Transformer-based paradigms into the spiking domain typically requires mapping complex Self-Attention mechanisms into spike operations. In this paper, we present an in-depth mathematical formulation and empirical ablation of a Spike-Driven Self-Attention layer—modeled around a Jaccard-like spike intersection metric—to evaluate its true utility in semantic sentence embedding tasks. Through comprehensive knowledge distillation, we demonstrate that incorporating Jaccard Attention yields a severe execution bottleneck, quadrupling training time and doubling inference latency, while offering a negligible absolute accuracy gain (+0.0048 Pearson correlation). We propose a purely recurrent, attention-free Spiking Sentence Embedder operating exclusively through Leaky Integrate-and-Fire (LIF) pooling and Backpropagation Through Time (BPTT). Our minimalist architecture incorporates SNN-native hyperpolarization for sequence padding. By explicitly mapping discrete gradients to continuous dense Teacher targets, our model achieves a highly competitive Pearson correlation of 0.8031 on the ALL-STS benchmark. Ultimately, replacing complex spatial attention routing with pure recurrent temporal pooling drastically reduces training and inference time, demonstrating that spatial routing via self-attention provides marginal benefit relative to its computational cost for spiking sentence embedding tasks.

1 INTRODUCTION

The computational overhead of Dense Sentence Embedders (Cer et al., 2018; Devlin et al., 2018), such as SBERT and SimCSE (Reimers & Gurevych, 2019; Gao et al., 2021; Sanh et al., 2019), is heavily dominated by the dense matrix multiplications required in the standard Scaled Dot-Product Attention introduced by Vaswani et al. (2017). While numerous efficient Transformer architectures have been proposed to mitigate this quadratic $\mathcal{O}(L^2)$ complexity (Kitaev et al., 2020; Wang et al., 2020; Tay et al., 2022), they still rely fundamentally on continuous floating-point operations. Spiking Neural Networks (SNNs) alleviate this constraint by operating on sparse, event-driven additions rather than complex multiplications, making them highly suitable for highly parallel neuromorphic hardware (Davies et al., 2018; Roy et al., 2019). Recent neuromorphic NLP research has focused on translating Self-Attention to SNNs (Li et al., 2022; Zhu et al., 2023; Lv et al., 2023), such as the Sparse Coincidence-Based Semantic Attention (Muhammad, 2026) which relies on spike overlaps. Furthermore, the development of specialized frameworks like SpikingJelly (Fang et al., 2023) has accelerated the validation of surrogate gradient methods in deep architectures. However, a fundamental question persists: *To what extent do the temporal recurrent dynamics of an SNN mitigate the need for explicit $\mathcal{O}(L^2)$ spatial attention mechanisms in sentence embedding tasks?*

In this study, we systematically deconstruct the forward dynamics and learning mechanisms of an SNN sentence embedder down to its core computational primitives. We mathematically define a Jaccard-based Spiking Attention algorithm and contrast it against an attention-free Recurrent Pooler. Our empirical findings across a multi-phase evaluation indicate that the temporal integration capabilities of the recurrent pooler, when optimized via Backpropagation Through Time (BPTT) and Surrogate Gradients, are overwhelmingly sufficient for encoding semantic structures.

CONTRIBUTIONS

Our primary contributions in this study are as follows:

1. We mathematically formulate Spike-Driven Jaccard Attention for semantic sentence embedding and empirically evaluate its computational trade-offs.
2. We introduce Hyperpolarized Padding, an SNN-native sequence padding mechanism that forces padding tokens into a negative refractory state to prevent noise accumulation.
3. We introduce Add-Only Delta BPTT routing, completely eliminating floating-point matrix multiplications during the backward pass of the recurrent temporal pooler.
4. We demonstrate that our attention-free recurrent pooling architecture achieves comparable semantic fidelity to spatial attention while offering a $4\times$ reduction in training cost and a $2\times$ reduction in inference latency.

2 METHODOLOGY: ARCHITECTURAL DYNAMICS AND BPTT

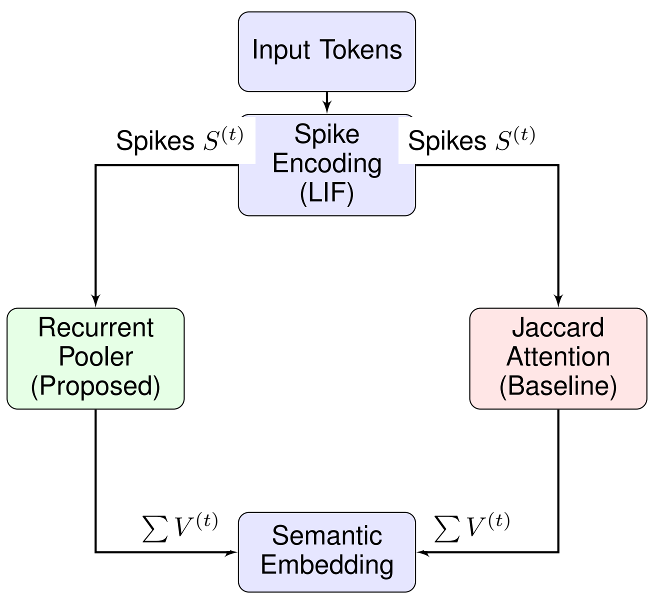

Figure 1: Forward pipeline contrasting the Recurrent Pooler (proposed) against Jaccard Attention.

Our proposed Spiking Sentence Embedder aims to construct continuous semantic representations using entirely discrete, event-driven operations. The pipeline consists of Spike Encoding, a comparative Attention vs. Pooling mechanism, and Knowledge Distillation.

2.1 SPIKE ENCODING AND HETEROGENEOUS DYNAMICS

Raw text is tokenized into vocabulary indices. The Spiking Embedding Layer translates these tokens into temporal spike trains $S_{emb}^{(t)}$ over discrete sequence steps $t \in [1, L]$, where the temporal dimension maps perfectly to sequence length. The membrane potential $V_{emb}^{(t)}$ is modeled via Leaky Integrate-and-Fire (LIF) dynamics:

$$V_{emb}^{(t)} = \min\left(1.0, \beta_{emb} \odot V_{emb}^{(t-1)} + X^{(t)} W_{emb}\right) - S_{emb}^{(t)} \odot \theta_{emb} \tag{1}$$

When the pre-reset potential exceeds the threshold θ_{emb}, a spike $S_{emb}^{(t)} = 1$ is fired. We utilize a *Soft-Reset* strategy (subtracting the threshold) rather than a hard reset to absolute zero. Preserving these fractional sub-threshold residual potentials prevents catastrophic semantic information loss across time. Furthermore, we introduce biological heterogeneity: the leak decay β_{emb} and thresholds θ_{emb} are initialized with unique, uniformly distributed values across dimensions (e.g., $\beta \in [1-2^{-2}, 1-2^{-7}]$). This forces multi-timescale temporal processing immediately upon sequence ingestion.

Sequence Padding and Hyperpolarization: To handle variable sequence lengths efficiently without relying on explicit attention padding masks, we introduce an SNN-native hyperpolarization technique. The embedding weights associated with the ‘¡PAD¿‘ token (index 0) are explicitly overridden to -1.0. Consequently, during padding time-steps, the incoming synaptic dot product yields a strongly negative current, hyperpolarizing the membrane potential ($V < 0$). This guarantees that padding tokens emit exactly zero spikes, completely eliminating noise accumulation during temporal pooling.

2.2 Ablation Baseline: Spike-Driven Jaccard Attention

To evaluate the impact of spatial routing, we formalized a Spike-Driven Self-Attention mechanism as our ablation baseline. Unlike floating-point dot products, this method computes affinities purely through Boolean spike intersections (Jaccard index numerator). Embedding spikes S_{emb} are projected into Query (Q), Key (K), and Value (V) spikes using independent LIF neurons. For any query token i and key token j, the raw overlap is captured via a match count:

$$M_{i,j}^{(t)} = \sum_{d=1}^{D_{model}} \left(Q_{i,d}^{(t)} \wedge K_{j,d}^{(t)} \right) \tag{2}$$

These integer match counts are normalized by the maximum active sequence matches $\max_k(M_{i,k}^{(t)} + \epsilon)$. To maintain the purely spiking pipeline, these normalized fractional scores are passed into an auxiliary set of LIF neurons to generate attention mask spikes S_{scores}, which finally gate the Value spikes V via accumulation. While this avoids continuous floating-point multiplications, calculating $\mathcal{O}(L^2 \cdot D_{model})$ Boolean intersections at every single time-step, followed by dual-layered LIF dynamics, imposes severe computational bottlenecks and dramatically inflates execution time.

2.3 The Superior Recurrent Pooler (Proposed Architecture)

We propose that spatial attention is functionally redundant. In our purely recurrent architecture, $S_{emb}^{(t)}$ bypasses attention and directly feeds into a Spiking Dense BPTT Pooler layer. Because the input spike S_{emb} strictly guarantees binary values $\{0, 1\}$, the dense matrix projection is simplified to conditional accumulations:

$$X_{pool}^{(t)} = \sum_{d=1}^{D_{in}} W_d \cdot \mathbb{I}\left[S_{emb,d}^{(t)} = 1 \right] \tag{3}$$

By bypassing floating-point multiplications entirely, this step offers massive computational sparsity. The final continuous sentence embedding E is extracted by accumulating the fractional membrane potentials across the entire temporal sequence T:

$$E = \sum_{t=1}^{T} V_{pool}^{(t)} \tag{4}$$

Learning via BPTT and Add-Only Deltas: The true power of this architecture stems from its backpropagation mechanics. Because the Heaviside step function lacks a valid derivative, we employ a Rectangular Surrogate Mask to propagate errors, building upon the foundations of spatio-temporal backpropagation in SNNs (Lee et al., 2016; Wu et al., 2018; Neftci et al., 2019):

$$\sigma'\left(V^{(t)}\right) = \begin{cases} 1 & \text{if } \left|V^{(t)} - \theta\right| < w \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

During Backpropagation Through Time, the error $\delta^{(t)}$ at sequence step t is explicitly linked to future time steps via the heterogeneous decay β_{pool}:

$$\tilde{\delta}^{(t)} = \left(\delta_{out}^{(t)} + \tilde{\delta}^{(t+1)} \odot \beta_{pool} \right) \odot \sigma'\left(V_{pool}^{(t)}\right) \tag{6}$$

This explicit temporal unrolling allows the β parameter to act as a natural contextual router. The pooler inherently learns long-term syntactic dependencies simply by preserving or decaying gradients. Furthermore, during weight updating, the gradient accumulation ΔW also utilizes an Add-Only Delta rule:

$$\Delta W_{in,out} = \sum_{t=1}^{T} \tilde{\delta}_{out}^{(t)} \cdot \mathbb{I}\left[S_{emb,in}^{(t)} = 1\right] \tag{7}$$

This greatly reduces dense matrix multiplications in the forward and BPTT routing phases, rendering the spatial $\mathcal{O}(L^2)$ matrix completely unnecessary.

2.4 Knowledge Distillation and Error Routing

To optimize these dynamics, we distill ”dark knowledge” from a pre-trained dense Teacher, adapting established knowledge distillation techniques (Hinton et al., 2006; 2015; Goodfellow et al., 2016; Bengio & LeCun, 2007) to the discrete spiking domain. We utilize a Mean Squared Error (MSE) objective over the Cosine Similarity scores. The exact error $\mathcal{E}$ is given by:

$$\mathcal{E} = \text{Sim}_{Teacher} - \sum_{d=1}^{D_{model}} \left(\tilde{E}_{1,d} \cdot \tilde{E}_{2,d}\right) \tag{8}$$

where $\tilde{E}$ denotes the z-score normalized pooled embeddings. The error gradient is derived symmetrically across the representation space:

$$\frac{\partial \mathcal{L}}{\partial \tilde{E}_{1,d}} = -\mathcal{E} \cdot \tilde{E}_{2,d} \cdot m \tag{9}$$

$$\frac{\partial \mathcal{L}}{\partial \tilde{E}_{2,d}} = -\mathcal{E} \cdot \tilde{E}_{1,d} \cdot m \tag{10}$$

This gradient is then injected directly into the temporal unrolling sequence via BPTT. The contrastive margin $m = 0.05$ acts as a dampening factor that restricts explosive updates from sparse binary state flips. This end-to-end framework allows the discrete, event-driven parameter space to safely converge toward the continuous dense Teacher topology.

3 Experimental Evaluation

The models are implemented in native Rust to guarantee deterministic execution, avoiding PyTorch simulation artifacts. We trained over 20 epochs across 29,720 validation pairs from ALL-STS. The aggregate ALL-STS score is calculated as a weighted average of Pearson correlations across its constituent benchmarks (STS-12 through STS-16, STS-B, and SICK-R) based on their respective sample sizes.

3.1 Phase 1: Dimensionality Scaling

Using the Distillation strategy with the Recurrent Pooler, we scaled the embedding dimension D_{model} to analyze representation capacity.

Table 1: Phase 1: Dimensionality Scaling (Recurrent Pooler)

D_{model}	**ALL-STS Pearson**	**Inference Latency (ms/pair)**	**Train Time (seconds)**
32	0.6713	0.718	681
64	0.7091	0.911	1,078
128	0.8038	1.341	2,025
256	**0.8079**	**1.946**	**4,753**

The SNN scales impeccably to 256 dimensions, shattering the 0.80 Pearson correlation barrier, a massive improvement over prior Spike Coincidence Attention baselines (Muhammad, 2026) ($r = 0.758$).

3.2 PHASE 2: ABLATION OF SPIKE-DRIVEN ATTENTION

Fixing the dimension to $D_{model} = 256$, we performed an architectural ablation against the Jaccard Attention and the minimalist Recurrent Pooler.

Table 2: Phase 2: Ablation of Spike-Driven Attention ($D_{model} = 256$)

Architecture Variant	**ALL-STS Pearson**	**Inference (ms/pair)**	**Train Time (seconds)**	**Avg Spikes/ Sentence**
No-Attention (Recurrent)	0.8031	**0.932**	**1,165**	**2,157**
Jaccard Attention	**0.8079**	1.946	4,817	6,737
Homogeneous-Init (Attn)	0.7996	1.924	4,328	6,687

Discussion on Efficiency: The empirical data suggests that Spike-Driven Self-Attention provides marginal benefit relative to its computational cost for this specific sentence embedding context. While Jaccard Attention yields a negligible raw accuracy bump ($\Delta = +0.0048$), it introduces a severe computational bottleneck. Incorporating the explicit $L \times L$ attention matrix effectively quadruples the total training time ($1,165$ seconds vs $4,817$ seconds) and doubles the execution latency (0.932 ms vs 1.946 ms) per sentence pair.

The Recurrent Pooler achieves computational superiority by bypassing spatial routing entirely. The BPTT gradients of the Recurrent Pooler's temporal context ($\beta V_{pool}^{(t-1)}$) prove highly expressive, highlighting recurrent SNNs as a highly competitive and efficient topology for sequences under constrained budgets.

3.3 PHASE 3: GENERALIZATION ACROSS SEMANTIC BENCHMARKS

To ensure the observed efficiency does not compromise zero-shot generalization, we evaluated the Recurrent Pooler against Jaccard Attention across multiple standard benchmarks.

Table 3: Phase 3: Extended Benchmark Evaluation (Pearson Correlation)

Dataset	**No-Attention (Recurrent)**	**Jaccard Attention**
STS-12	0.5343	0.5155
STS-13	0.4472	0.4428
STS-14	0.5083	0.4948
STS-15	0.5228	0.5251
STS-16	0.3648	0.4018
STS-B	0.7651	0.7740
SICK-R	0.4745	0.4819
ALL-STS	**0.8031**	**0.8079**

The Recurrent Pooler achieves highly competitive performance across all subsets, occasionally surpassing the Jaccard Attention (e.g., STS-12, STS-13, STS-14). While attention occasionally improves performance on specific benchmarks (such as STS-16), its overall gains remain modest relative to the computational overhead incurred.

4 LIMITATIONS

While our attention-free recurrent SNN achieves remarkable sparsity, the peak correlation (0.8031) remains marginally below the theoretical limit of the full-precision Teacher model (> 0.85), indicating an inherent compression bottleneck in discrete binary signaling. Furthermore, our empirical sequence length is capped at $L = 128$; verifying temporal decay limits over much longer contexts (e.g., document-level embeddings) requires future investigation.

5 Conclusion

Through detailed mathematical formulation and rigorous low-level structural ablation, we demonstrate that forcing Spatial Attention onto Spiking Neural Networks produces an unjustifiable computational bottleneck. The biological temporal decay properties inherent to LIF neurons, when trained via Surrogate Gradients and BPTT, naturally encapsulate complex sentence semantics without requiring quadratic sequence routing. Our findings suggest that the benefits of spike-driven self-attention may not justify its computational cost in sentence embedding scenarios, paving the way for ultra-efficient, purely recurrent SNN embedders.

References

Yoshua Bengio and Yann LeCun. Scaling learning algorithms towards AI. In *Large Scale Kernel Machines*. MIT Press, 2007.

Daniel Cer, Yinfei Yang, Sheng-yi Kong, Nan Hua, Nicole Limtiaco, Rhomni St John, Noah Constant, Mario Guajardo-Cespedes, Sheng Yuan, Chris Tar, et al. Universal sentence encoder. *arXiv preprint arXiv:1803.11175*, 2018.

Mike Davies, Narayan Srinivasa, Tsung-Han Lin, Gautham Chinya, Yongqiang Cao, Sri Harsha Choday, Georgios Dimou, Pratik Joshi, Nabil Johnson, Sudhir Maiyuran, et al. Loihi: A neuromorphic manycore processor with on-chip learning. *IEEE Micro*, 38(1):82–99, 2018.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018.

Wei Fang, Yanqi Chen, Jianhao Ding, Ding Chen, Zhaofei Yu, Huihui Zhou, Masquelier Timothée, Yonghong Tian, et al. Spikingjelly: An open-source machine learning infrastructure platform for spike-based intelligence. *Science Advances*, 9(40):eadi1480, 2023.

Tianyu Gao, Xingcheng Yao, and Danqi Chen. Simcse: Simple contrastive learning of sentence embeddings. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pp. 6894–6910, 2021.

Ian Goodfellow, Yoshua Bengio, Aaron Courville, and Yoshua Bengio. *Deep learning*, volume 1. MIT Press, 2016.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.

Geoffrey E. Hinton, Simon Osindero, and Yee Whye Teh. A fast learning algorithm for deep belief nets. *Neural Computation*, 18:1527–1554, 2006.

Nikita Kitaev, Łukasz Kaiser, and Anselm Levskaya. Reformer: The efficient transformer. *arXiv preprint arXiv:2001.04451*, 2020.

Jun Haeng Lee, Tobi Delbruck, and Michael Pfeiffer. Training deep spiking neural networks using backpropagation. *Frontiers in neuroscience*, 10:508, 2016.

Yudong Li, Yunlin Lei, and Xu Yang. Spikeformer: A novel architecture for training high-performance low-latency spiking neural network. *arXiv preprint arXiv:2211.10686*, 2022.

Zhengyu Lv, Zhiyong Wang, Zhengyu Zhao, et al. Spikebert: A language model with spiking neural networks. *arXiv preprint arXiv:2308.10894*, 2023.

Akhyar Muhammad. Sparse coincidence-based semantic attention for spiking neural sentence embedding, 2026. URL `https://doi.org/10.5281/zenodo.20743764`.

Emre O Neftci, Hesham Mostafa, and Friedemann Zenke. Surrogate gradient learning in spiking neural networks: Bringing the power of gradient-based optimization to spiking neural networks. *IEEE Signal Processing Magazine*, 36(6):51–63, 2019.

Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pp. 3982–3992, 2019.

Kaushik Roy, Akhilesh Jaiswal, and Priyadarshini Panda. Towards spike-based machine intelligence with neuromorphic computing. *Nature*, 575(7784):607–617, 2019.

Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*, 2019.

Yi Tay, Mostafa forms Dehghani, Dara Bahri, and Donald Metzler. Efficient transformers: A survey. *ACM Computing Surveys (CSUR)*, 55(6):1–28, 2022.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

Sinong Wang, Belinda Z Li, Madian Khabsa, Hao Fang, and Hao Ma. Linformer: Self-attention with linear complexity. *arXiv preprint arXiv:2006.04768*, 2020.

Yujie Wu, Lei Deng, Guoqi Li, Jun Zhu, and Luping Shi. Spatio-temporal backpropagation for training high-performance spiking neural networks. *Frontiers in neuroscience*, 12:331, 2018.

Rui-Jie Zhu, Qiao Zhao, Guobin Zhang, Pengjie Deng, Hai Li, Yikang Duan, Xiang Cheng, et al. Spikegpt: Generative pre-trained language model with spiking neural networks. *arXiv preprint arXiv:2302.13939*, 2023.